

VLAN (Virtual Local Area Network)

Definition

- A **VLAN (Virtual Local Area Network)** is a **logical subdivision** of a physical network.
- It allows a single switch to be **logically divided into multiple smaller networks**, each acting as an independent LAN.
- Devices within the same VLAN can communicate directly, while communication between different VLANs requires a **router or Layer 3 switch**.

Purpose

- To **segregate network traffic** for better performance, security, and management.
- To group users by **function or department**, not by physical location.
- To **reduce broadcast domains** — each VLAN acts as a separate broadcast domain.

Key Concepts

Term	Explanation
Broadcast Domain	VLAN limits broadcast traffic within itself.
Trunk Port	Carries multiple VLANs between switches using tagging.
Access Port	Connects end devices (like PCs) to a single VLAN.
VLAN ID	Unique number assigned to identify a VLAN (1–4094).
Inter-VLAN Routing	Communication between VLANs through a router or Layer 3 switch.

Advantages

1. **Improved Security:** Traffic from one VLAN is isolated from others.
2. **Better Performance:** Reduces unnecessary broadcast traffic.
3. **Flexibility:** Users can be grouped logically regardless of location.
4. **Simplified Management:** Easier network configuration and maintenance.
5. **Efficient Resource Use:** Segments a large network into smaller, manageable parts.

Disadvantages

1. **Complex Configuration:** Requires understanding of tagging and trunking.

2. **Inter-VLAN Routing Needed:** Communication between VLANs requires extra configuration.
 3. **Misconfiguration Risks:** Wrong VLAN assignments can cause connectivity issues.
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Types of VLANs

1. **Default VLAN:** VLAN 1 – all switch ports belong to it by default.
 2. **Data VLAN:** Used for user-generated data (e.g., VLAN 10 for HR).
 3. **Voice VLAN:** Used for IP phone traffic.
 4. **Management VLAN:** Used for switch management access (e.g., SSH, Telnet).
 5. **Native VLAN:** Used for untagged traffic on trunk links.
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Implementation of VLAN with Wireless Router Connection

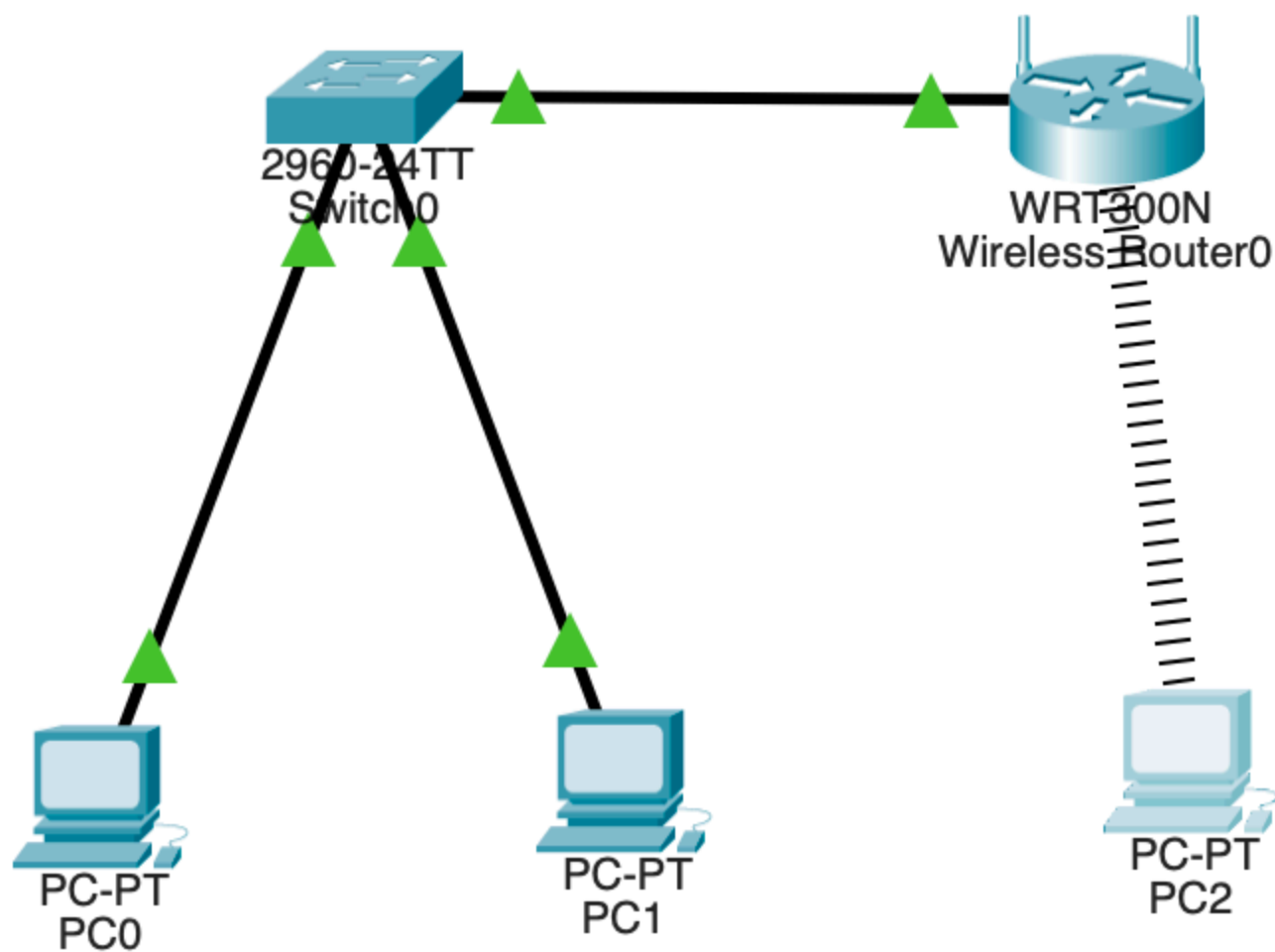
Aim

To configure VLANs on a switch and integrate a wireless router to allow wireless connectivity in one VLAN, while maintaining VLAN isolation between wired networks.

Apparatus / Tools

- Cisco Packet Tracer
- 1 × 2960-24TT Switch
- 1 × Wireless Router (WRT300N or similar)
- 2 × PCs (wired)

- 1 × Laptop (with WPC300N wireless module)



IP Addressing Table

Device	Interface	VLAN	IP Address	Subnet Mask	Gateway
Wireless Router0	LAN	10	192.168.10.1	255.255.255.0	—
PC0	FastEthernet0	10	DHCP	255.255.255.0	192.168.10.1
PC1	FastEthernet0	20	192.168.20.2	255.255.255.0	192.168.20.1
Laptop	Wireless	10	DHCP	255.255.255.0	192.168.10.1

Step-by-Step Configuration

1. Switch Configuration

Open **Switch0** → **CLI** and enter:

```
enable
configure terminal
```

```
vlan 10
  name Admin
vlan 20
  name Sales
exit
```

Assign VLANs to switch ports:

```
interface fastEthernet0/1
  switchport mode access
  switchport access vlan 10
exit

interface fastEthernet0/2
  switchport mode access
  switchport access vlan 20
exit

interface fastEthernet0/3
  switchport mode access
  switchport access vlan 10
exit

end
write
```

Connections:

- PC0 → F0/1
- PC1 → F0/2
- Wireless Router (LAN port) → F0/3

2. Wireless Router Configuration

Open **Wireless Router0** → **Config Tab**

A. Internet Tab

- Select: **DHCP** (automatic)

B. LAN Tab

IP Address: 192.168.10.1
Subnet Mask: 255.255.255.0

C. Wireless Tab

SSID: VLAN10-WiFi
Authentication: WPA2-PSK
PSK Pass Phrase: 12345678

D. DHCP Configuration (GUI Tab → Network Setup)

DHCP Server: Enabled
Start IP Address: 192.168.10.2
Maximum Users: 50
Static DNS: 8.8.8.8

3. End Devices Configuration

PC0 (VLAN 10)

- Connection: Switch F0/1
- IP Configuration: **DHCP**

PC1 (VLAN 20)

- Connection: Switch F0/2
- IP Configuration: **Static**

IP: 192.168.20.2
Subnet Mask: 255.255.255.0
Gateway: 192.168.20.1

Laptop (VLAN 10 – Wireless)

- Add **WPC300N** module in the **Physical** tab.
- Go to **Desktop** → **PC Wireless** → **Connect**

SSID: VLAN10-WiFi
Password: 12345678

- IP Configuration: **DHCP**
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4. Verification Commands

A. Check VLAN Configuration (on Switch CLI)

```
show vlan brief
```

Expected:

VLAN10 – Fa0/1, Fa0/3
VLAN20 – Fa0/2

B. Check IP Addresses (on PCs)

```
ipconfig
```

Confirms IP assignment by DHCP or static.

C. Ping Tests

Test	Command	Expected Result
PC0 → Router	ping 192.168.10.1	Successful
Laptop → Router	ping 192.168.10.1	Successful
PC0 → Laptop	ping 192.168.10.3 (DHCP IP)	Successful
PC1 → PC0	ping 192.168.10.x	Request timed out (different VLANs)

5. Results

- VLAN 10 (wired + wireless) devices communicate successfully.
- VLAN 20 remains isolated.
- DHCP works via Wireless Router.
- Wireless clients get connected using SSID VLAN10-WiFi.